Rotation Sweep Findings

IYPT Pinhole Sunglasses Project — Array Design Sub-Study

August 16, 2026

1 The question this task answers

Does any rotated square/hexagonal/rectangular candidate beat `jittered_lattice`'s current winning density among structurally-safe survivors, once lattice orientation (relative to this project's own established anisotropic-gaze finding) is actually tested instead of assumed fixed at one arbitrary angle?

2 Headline result

No – and the reversal is total, not marginal: `jittered_lattice` now wins at ZERO of 9 M-scopes (it previously won 5 of 9: $M = 1, 2, 3, 6, 8$). A **rotated** `square` or `hexagonal` lattice now wins at 6 of 9 scopes ($M = 1, 2, 3, 6, 8$, global); `archimedean_spiral` still wins $M = 4/5$; `rectangular` still wins $M = 0$. **This is a real, reportable change to the Metric-1 winner list, not a rounding effect** – confirmed by direct inspection of the underlying gate values, not just the summary density numbers (see Section 5).

3 Per-M comparison: old winner vs. new (rotation-inclusive) winner

M	old density (um^{-2})	new winner	new density (um^{-2})	change
0.0	2.9630e-07	rectangular (rot=0)	2.9630e-07	same winner
1.0	9.4864e-08	**square (rot=45°)**	1.1111e-07	**CHANGED, +17.1%**
2.0	1.0126e-07	**square (rot=45°)**	1.1111e-07	**CHANGED, +9.7%**
3.0	1.0126e-07	**square (rot=45°)**	1.1111e-07	**CHANGED, +9.7%**
4.0	2.3085e-07	archimedean_spiral	2.3085e-07	same winner
5.0	2.3085e-07	archimedean_spiral	2.3085e-07	same winner
6.0	1.0471e-07	**square (rot=45°)**	1.1111e-07	**CHANGED, +6.1%**
8.0	1.0614e-07	**square (rot=45°)**	1.1111e-07	**CHANGED, +4.7%**
global*	2.5000e-07	**hexagonal (rot=30°)**	2.8868e-07	**CHANGED, +15.5%**

Table 1: Old winner column omits family (all 5 changed rows were `jittered_lattice`, except global which was `square` at rot=0). *global's winner-change is flagged with a numerical-fragility caveat, see Section 5. Full per-row detail: `comparison_metric/metric1_max_density_per_M.csv` (new) vs. `metric1_max_density_per_M_PREROTATION_legacy.csv` (old).

4 How far behind is `jittered_lattice` now, even where it lost?

Comparing the new winner's density against `jittered_lattice`'s OWN best structurally-safe density at each M (not just "did it win," but "by how much"):

M	**new winner leads jittered_lattice's own best by**
0.0	+15.1%
1.0	+17.1%
2.0	+9.7%
3.0	+9.7%
4.0	+123.5%
5.0	+123.5%
6.0	+6.1%
8.0	+4.7%
global	+122.7%

`jittered_lattice` is not close anywhere – the smallest margin ($M = 8$) is still a 4.7% lead for the new winner, and at $M = 4/5$/global the gap is over 100% (`archimedean_spiral`/`hexagonal` were always well ahead there; rotation only changed who's ahead at the other 6 scopes).

5 Mechanism – confirmed by direct inspection, not inferred from the summary alone

Density itself never changes with rotation (rotating a rigid lattice changes neither its packing density nor its ghost margin, which depends only on pairwise distance). What changes is **which candidates pass the safety gates** – specifically `worst_case_vignetting_margin`, exactly as this task's own premise predicted (orientation relative to the anisotropic gaze axis matters for vignetting, not for ghosting).

$M = 1, 2, 3, 6, 8$ (square, pitch=3000um, $N = 50$) – clean, robust, unaffected by any numerical edge case

Direct check of the underlying rows:

rotation_deg	worst_case_vignetting_margin	structurally_safe
0.0	-5.939 / -7.050 (varies by M)	False
15.0	-7.050	False
30.0	-7.050	False
45.0	$+\infty$	**True** ← flips to safe

`N_actual` for this candidate is **perfectly stable across all 4 rotation angles** (20/20/20/20, 50/50/50/50, 100/100/100/100 – verified directly) – this pitch (3000um) is far from the hole diameter (a_hole=1250um at $M = 1$D, ratio=0.417), so no overlap-rejection boundary is anywhere near being touched. This is a genuine geometric effect: rotating the lattice by 45 degrees moves specific hole positions enough that no pair of holes is ever simultaneously active under the pupil across the full gaze sweep (`worst_case_vignetting_margin` goes to a clean $+\infty$, not a marginal small-positive value), where at 0/15/30 degrees a pair does become active at some tested gaze angle and produces a firmly negative margin. **High confidence finding.**

$M =$global (hexagonal, pitch=2000um, rotation=30°) – real, but flagged as numerically fragile

Here `pitch_um` (2000) exactly equals `a_hole_um` (2000, the largest acuity-optimal aperture across the M range, used for global-scope overlap validity per this project's own established convention) – **exact tangency, ratio=1.0**, a case this project's own `compute_density.py` already documents as fragile: `reject_overlapping()`'s strict $d^2 > a_{\text{hole}}^2$ test sits exactly on a floating-point knife-edge at exact tangency. Direct check confirms `N_actual` for this candidate actually VARIES across

rotation angles at the same $N_{\text{requested}}$ (e.g. $N_{\text{requested}} = 20$ gives $N_{\text{actual}} = 7, 9, 7, 9$ at rotation= 0/15/30/45 respectively) – a symptom of rotation's own floating-point rounding (irrational cos/sin values) nudging which near-tied points survive the exact-tangency boundary, not a bug in the rotation transform itself (verified elsewhere that rotation preserves `N_actual` exactly away from this boundary – see Section 6).

The M =global winner-change is real for the specific rotation angles tested (rotation=30° genuinely reaches `structurally_safe=True` with `worst_case_vignetting_margin`= $+\infty$ at this pitch), **but it should be read as "a numerically narrow window this project's own known exact-tangency edge case opened up," not as a broad, robustly-safe orientation** the way the $M = 1$–8 finding is. Recommend re-checking this specific candidate at a pitch slightly off exact tangency (e.g. 1990 or 2010um) before treating it as a settled design choice.

6 Step 1 order requirement: rotation_deg=0.0 exact-reproduction, confirmed before the sweep ran

Per this task's explicit ORDER requirement, `spot_check_generators.py` was extended and run BEFORE the rotation sweep: `rotation_deg=0.0` reproduces the pre-rotation-support output bit-for-bit (`np.array_equal`, not just `np.allclose`) for all three families, and `rotation_deg=30.0` was independently verified to (a) actually change coordinates, (b) preserve `N_actual` exactly, and (c) preserve every pairwise distance exactly – all PASS. (The one place `N_actual` does vary with rotation in the real sweep, described above, is the pre-existing exact-tangency edge case, confirmed by direct inspection to be a floating-point boundary effect specific to `pitch_um==a_hole_um`, not a contradiction of this check – the spot-check's own $N = 50$/pitch=900/a_hole=400 case is nowhere near that boundary.)

7 What was changed, and what was preserved

- `generators/base_generator.py`: added `rotate_points(coords_um, rotation_deg)` (shared helper, avoids duplicating the same rotation matrix in 3 files). `reject_overlapping`, `recenter_on_centroid`, `finalize` unchanged.
- `generators/square.py`, `generators/hexagonal.py`, `generators/rectangular.py`: added `rotation_deg=0.0` parameter, applied before `bg.finalize()`. Core lattice construction (meshgrid, radial sort) unchanged.
- `sweep/run_rotation_sweep.py` (new): re-swept ONLY square/hexagonal/rectangular, ONLY adding rotation_deg in $\{0, 15, 30, 45\}$ on the EXISTING N/spacing grid – no other family touched or re-run.
- `sweep_results.csv`: original preserved as `sweep_results_PREROTATION_legacy.csv` (6395 rows, untouched). Existing square/hexagonal/rectangular rows retroactively tagged with `;rotation_deg=0.0` in their `spacing_params` string (text-only, no numeric column touched). New rows for rotation in $\{15, 30, 45\}$ appended. **No duplicate rows** – verified by exact-duplicate count matching the number of existing tagged rows precisely (702), not assumed; a first attempt at this check found a real bug (an `M`-column string/float dtype mismatch from the CSV round-trip silently broke `drop_duplicates()`, caught because the dedup count didn't match the expected value and was investigated before proceeding). Final: 8933 rows.
- `sweep_results_with_density.csv`: re-run via the UNMODIFIED `degeneracy_filter.py`/`compute_density.py`. Previous version preserved as `sweep_results_with_density_PREROTATION_legacy.csv`. 432 rows excluded (108 designs x 4 rotation angles – the same known-degenerate combinations recurring per rotation, since the degeneracy

criteria are rotation-invariant). `open_area_fraction<=1.0001` assertion: **passed, zero violations**, 8501 surviving rows.

- `comparison_metric/metric1_max_density_per_M.csv`: re-run via the UNMODIFIED `gates.py/metrics.py` (Metric 1 only, via the new `comparison_metric/run_metric1_only.py`). Previous version preserved as `metric1_max_density_per_M_PREROTATION_legacy.csv`.
- Untouched, as required: every other family's data (concentric, fibonacci_spiral, golden_angle, archimedean_spiral, ring_offset_concentric, poisson_disk, jittered_lattice), `array_optimization/`, and every Phase 2 confirmation file.

8 Reproduction

```
cd Array_Design/array_geometry_lab
python3 spot_check_generators.py              # step 1 gate -- must pass first
python3 sweep/run_rotation_sweep.py            # step 2-3
cd comparison_metric
python3 compute_density.py                     # step 4
python3 run_metric1_only.py                    # step 5
```